INCH POUND
MIL-PRF-1/272F
20 May 2014
SUPERSEDING
MIL-PRF-1/272E
26 June 1998

PERFORMANCE SPECIFICATION SHEET

# ELECTRON TUBE, CATHODE RAY
# TYPES 2BP1 AND 2BP11

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Electrostatic deflection and focus.

DIMENSIONS AND PIN CONNECTIONS: See figure 1.

ABSOLUTE RATINGS:

| Parameter: | Type phosphor | Ef | Ec1 | ed | Eb1 | Eb2 | Light output | Rg | Zd | Ehk | Alt |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | | V | V dc | v | V dc | V dc | ftL | MegΩ | MegΩ | V dc | ft |
| Maximum: | P1, P11 | 6.9 | -200 | 550 | 1,100 | 2,750 | --- | 1.5 | 1.0 | ± 125 | 30,000 |
| Minimum: | P1 | 5.7 | 0 | --- | --- | 500 Note 2 | 4.0 | --- | --- | --- | --- |
| Minimum: | P11 | 5.7 | 0 | --- | --- | 1,100 | 2.25 | --- | --- | --- | --- |
| Test conditions: | P1, P11 | 6.3 | Adj | --- | Focus | 1,000 | --- | --- | --- | --- | --- |

GENERAL:

Qualification - Required (see Note 1 at end of Table I).

This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:08Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and inspection.

| Inspection | Method MIL-STD-1311 | Types | Notes | Conditions | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Min | Max | |
| Qualification inspection | | | | | | | | |
| Vibration | 5111 | P1, P11 | --- | | Width | --- | 1.0 | mm |
| Barometric pressure, reduced | 1002 | P1, P11 | 6 | | --- | --- | --- | --- |
| Neck and bulb alignment (electrostatic types) | 5101 | P1, P11 | --- | | Diameter | --- | 1.63 | Inch |
| Cathode illumination | 5216 | P1, P11 | --- | | --- | --- | --- | --- |
| Focusing voltage atmodulation condition | 5246 | P1, P11 | --- | | Eb1 | 150 | 280 | V dc |
| Deflection-factor uniformity | 5248 | P1, P11 | --- | | --- | --- | --- | --- |
| Direct-interelectrode capacitance | 1331 | P1, P11 | --- | g1 to all<br>D1 to D2<br>D3 to D4<br>D1 to all<br>D2 to all<br>D3 to all<br>D4 to all | Cg1<br>C1D2<br>C3D4<br>CD1<br>CD2<br>CD3<br>CD4 | ---<br>---<br>---<br>---<br>---<br>---<br>--- | 11.0<br>4.0<br>4.0<br>15.0<br>11.0<br>10.0<br>11.0 | pF<br>pF<br>pF<br>pF<br>pF<br>pF<br>pF |
| Pressure (implosion) | 1141 | P1, P11 | --- | | --- | --- | --- | --- |
| Base material insulating quality | 1216 | P1, P11 | --- | | --- | --- | --- | --- |
| Conformance inspection, part 1 | | | | | | | | |
| Voltage breakdown | 5201 | P1, P11 | --- | | --- | --- | --- | --- |
| Voltage breakdown (electrostatic types) | 5201 | P1, P11 | --- | | --- | --- | --- | --- |
| Gas “cross” | 5206 | P1<br>P11 | 7<br>7 | Light = 4.0 ftL<br>Light = 2.25 ftL | ---<br>--- | ---<br>--- | ---<br>--- | ---<br>--- |
| Bulb, screen, and faceplate quality | 5106 | P1, P11 | --- | | --- | --- | --- | --- |
| Light output | 5221 | P1<br>P11 | 7<br>5, 7 | | Light<br>Light | 4.0<br>2.25 | ---<br>--- | ftL<br>ftL |
| Modulation | 5223 | P1<br>P11 | ---<br>--- | Light = 4.0 ftL<br>Light = 2.25 ftL | $\Delta$Ec1<br>$\Delta$Ec1 | ---<br>--- | 48<br>48 | V dc<br>V dc |
| Spot position (electrostatic deflection) | 5231 | P1, P11 | --- | | --- | --- | 10 | mm |
| Spot displacement (leakage) | 5231 | P1, P11 | --- | | Displ | --- | 6 | mm |
| Grid cutoff voltage | 5241 | P1, P11 | --- | | Ec1 | --- | -67.5 | V dc |

See notes at end of Table I.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:08Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and inspection - Continued.

| Inspection | Method MIL-STD-1311 | Types | Notes | Conditions | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Min | Max | |
| Conformance inspection, part 2 | | | | | | | | |
| Heater current | 1301 | P1, P11 | --- | | If | 540 | 660 | mA |
| Electrode current (anode No. 1) | 5201 | P1, P11 | --- | Ec1 = 0 | Ib1 | -15 | 10 | µA dc |
| Electrode current (cathode) | 5201 | P1<br>P11 | ---<br>--- | Light = 4.0 ftL<br>Light = 2.25 ftL | Ik<br>Ik | ---<br>--- | 1,000<br>1,000 | µA dc<br>µA dc |
| Base alignment (electrostatic types) | 5101 | P1, P11 | --- | 3D4; pin No. 1 | --- | --- | --- | --- |
| Angle between traces | 5101 | P1, P11 | --- | | Angle | 87 | 93 | Degrees |
| Neck and base alignment (electrostatic types) | 5101 | P1, P11 | --- | | --- | --- | --- | --- |
| Stray light emission | 5216 | P1, P11 | --- | | --- | --- | --- | --- |
| Line width “A” (electrostatic types) | 5226 | P1<br>P11 | 3<br>--- | Light = 4.0 ftL<br>Light = 2.25 ftL | Width<br>Width | ---<br>--- | 0.45<br>0.45 | mm<br>mm |
| Line width “B” (electrostatic deflection) | 5226 | P1<br>P11 | 4<br>4 | Light = 4.0 ftL<br>Light = 2.25 ftL | Width<br>Width | ---<br>--- | 0.45<br>0.45 | mm<br>mm |
| Focusing voltage | --- | P1<br>P11 | ---<br>--- | Light = 4.0 ftL<br>Light = 2.25 ftL | Eb1<br>Eb1 | 150<br>150 | 280<br>280 | V dc<br>V dc |
| Deflection factor (1D2) | 5248 | P1,P11 | --- | | DF | 115 | 155 | V dc/in. |
| Base pin solder depth | 1111 | P1, P11 | --- | | --- | --- | --- | --- |
| Deflection factor (3D4) | 5248 | P1, P11 | --- | | DF | 74 | 100 | V dc/in. |
| Heater-cathode leakage current | 5251 | P1, P11 | --- | | --- | --- | --- | --- |
| Grid No. 1 leakage current | 5251 | P1, P11 | --- | | --- | --- | --- | --- |
| Anode No. 2 leakage current | 5251 | P1, P11 | --- | | --- | --- | --- | --- |
| Secureness of base, cap, or insert | 1101 | P1, P11 | --- | | --- | --- | --- | --- |
| Permanence of marking | 1105 | P1, P11 | --- | | --- | --- | --- | --- |

See notes at end of Table I.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:08Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and inspection - Continued.

| Inspection | Method MIL-STD-1311 | Types | Notes | Conditions | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Min | Max | |
| Conformance inspection, part 3 | | | | | | | | |
| Life test | --- | P1 | --- | Group C; t = 500 hours (min); light = 4.0 ftL; Eb2 = 2,750 V dc | --- | --- | --- | --- |
| | | P11 | --- | Group C; t = 500 hours (min); light = 2.25 ftL; Eb2 = 2,750 V dc | --- | --- | --- | --- |
| Life-test end points: | | | | | | | | |
| Line width “A” | 5226 | P1 | --- | Light = 3.4 ftL | Width | --- | 0.45 | mm |
| Line width “B” | 5226 | P1 | --- | Light = 3.4 ftL | Width | --- | 0.45 | mm |
| Modulation | 5223 | P1 | --- | Light = 3.4 ftL | ΔEc | --- | 48 | V dc |
| Life-test end points: | | | | | | | | |
| Line width “A” | 5226 | P11 | --- | Light = 2.25 ftL | Width | --- | 0.45 | mm |
| Line width “B” | 5226 | P11 | --- | Light = 2.25 ftL | Width | --- | 0.45 | mm |
| Modulation | 5223 | P11 | --- | Light = 2.25 ftL | ΔEc | --- | 48 | V dc |

1. The construction of this gun shall be of the zero Ib1 type. The following information and material are to be forwarded with the four regular samples when application for qualification approval is made:

   1 each gun drawings with significant dimensions.
   1 each sample of gun to be used in manufacture of tubes.

2. This value is recommended only for low-velocity deflecting and low-ambient light levels.

3. The deflecting plates shall be returned to anode No. 2 through a minimum of 2.5 mΩ resistors. Ib3, the beam current, shall be set at 50 μA dc. The high-frequency scanning shall be applied to the deflecting plates nearest the screen and the amplitude shall be adjusted to give a line length of approximately 90 percent of the maximum tube diameter. The low-frequency scanning amplitude shall be expanded to approximately 90 percent of the maximum tube diameter in the direction perpendicular to the direction of high-frequency scanning. Readjustment may be made for best overall focus. The tube shall be observed for deflection defocusing, astigmatism, or spot ellipticity observable to the eye as evidenced by fuzziness due to lack of sharpness of trace (usually around edges), bow-tying (irregular widths of any single line when observed at different points), bowing of trace other than that normally caused by curvature of bulb. This test for focus is to be made in addition to the line width measurements.

4. The same conditions shall be set up as described in note 3 except that the connection of deflecting elements to the low- and high-frequency scanning supplies shall be interchanged and the amplitudes adjusted to 90 percent of the maximum tube diameter in both directions without any adjustment of focus from conditions in note 3. An examination for defocusing, astigmatism, or spot ellipticity shall be made as in note 3.

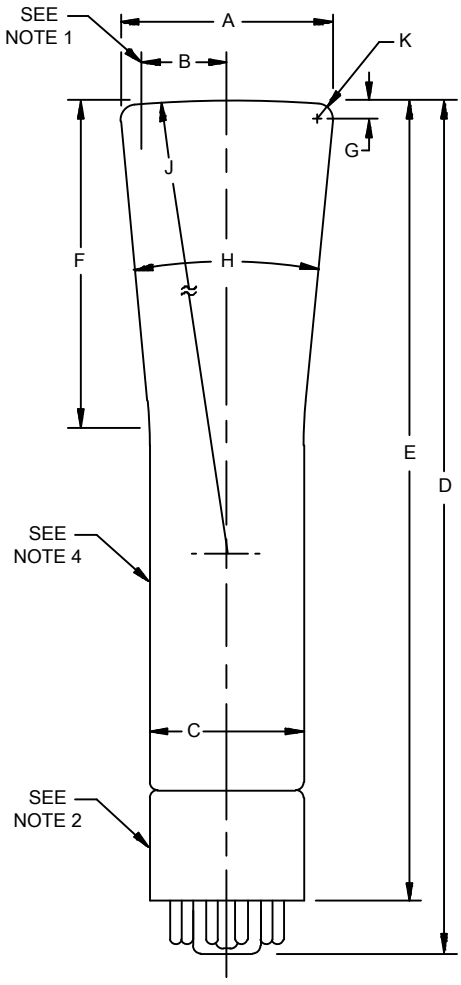
5. As measured by a 2 x 2-inch (50.8 x 50.8 mm) raster using a type 3 photronic cell without eye correction, calibrated in footcandles of illumination from a light source having color temperature of 2,700°K.

6. The test is made with maximum voltage (Egl maximum negative voltage) applied to the base pins, and deflection electrodes if applicable, only; and pressure of 30,000 feet (225 mmHg). Connections should be made in a manner that does not degrade the tube’s electrical voltage breakdown characteristics. Satisfactory operation is the absence of arc-over and corona.

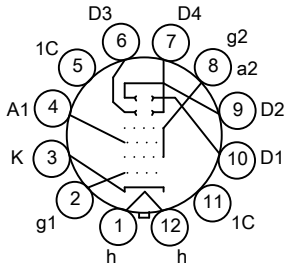
7. This test to be performed at the conclusion of the holding period.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:08Z
Check the source to verify that this is the current version before use.




| Ltr | Minimum | | Maximum | |
|---|---|---|---|---|
| | in | mm | in | mm |
| Conformance inspection, part 2 | | | | |
| A | 1.938 | 49.23 | 2.063 | 52.40 |
| B | 0.875 | 22.23 | | |
| C | 1.312 | 33.43 | 1.438 | 36.53 |
| D | 7.438 | 188.93 | 7.812 | 198.42 |
| E | 6.938 | 176.23 | 7.312 | 185.72 |
| Reference dimensions (see note 3) | | | | |
| | in | | mm | |
| F | 3.062 | | 77.77 | |
| G | 0.225 | | 5.72 | |
| H | 12° 37’ | | | |
| J | 8.00 R | | | |
| K | 0.188 R | | 4.78 R | |




NOTES:

1. The minimum useful screen radius.
2. The base shall be a small shell duodecal 12 pin or small shell duodecal minus pins 5 and 11 (JEDEC B10-75).
3. Reference dimensions are for information only and are not required for inspection purposes.
4. The bulb shall be a J16A type. Either clear or filter glass may be used.

FIGURE 1. Outline drawing of electron tube types 2BP1, 2BP11.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:08Z
Check the source to verify that this is the current version before use.

Referenced documents. In addition to MIL-PRF-1, this specification sheet references MIL-STD-1311.

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity :
DLA - CC

(Project 5960-2014-003)

Review activities :
Army - AR, MI
Navy - AS, CG, MC, OS
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:08Z
Check the source to verify that this is the current version before use.